■■ AEGIS-SOC: Technical Architecture & Data Flow Guide

Enterprise Autonomous Cyber Defense Operations Center // Powered by LangGraph Multi-Agent Swarm, Gemini 2.5 LLM & ChromaDB RAG

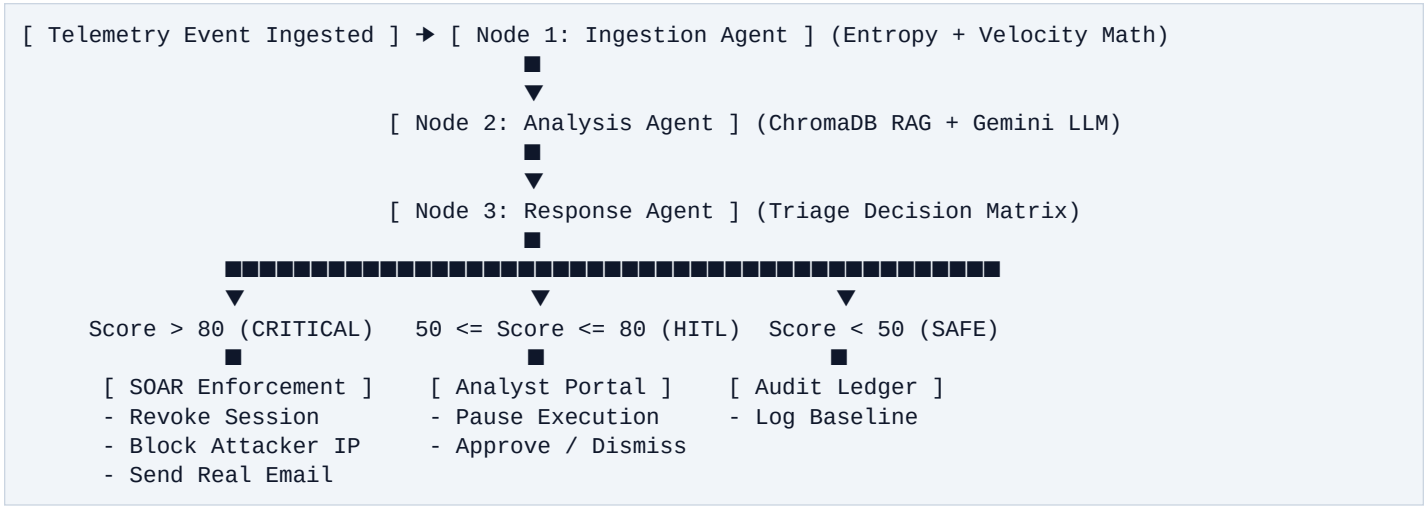
Executive Summary

AEGIS-SOC is an AI-driven, multi-agent autonomous security framework designed to detect, analyze, and mitigate advanced cyberattacks in real-time. By moving beyond traditional static firewall rules, AEGIS-SOC combines **mathematical anomaly engines** (Shannon Entropy & Haversine Travel Velocity) with **vector retrieval-augmented generation (RAG)** and **Google Gemini 2.5 LLM reasoning** to achieve explainable, low-latency incident response.

1. System Architecture Overview

The system is structured into 6 primary decoupled layers:
• **1. Data Ingestion Streamer:** O(1) generator reading 31M-row Kaggle RBA dataset.
• **2. FastAPI REST Gateway:** High-throughput API handling telemetry payloads & audit logs.
• **3. LangGraph Multi-Agent Swarm:** 3-node stateful pipeline (Ingestion ➔ Analysis ➔ Response).
• **4. Vector Memory & Cognitive AI:** Local ChromaDB vector database & Google Gemini 2.5 Flash LLM.
• **5. SOAR Mitigation Engine:** Session revocation, IP firewall blocks & real TLS email alerts.
• **6. Cyber Command HUD:** Interactive Sci-Fi HTML5 dashboard for real-time visualization.

2. End-to-End Telemetry Data Flow



3. Detailed Subsystem Component Breakdown

• server.py — FastAPI Gateway

Manages REST endpoints (/api/v1/telemetry/ingest, /api/v1/stats, /api/v1/rag/query), user trajectory tracking across locations, and sliding window 2,000-item audit ledger memory bounds.

- **agents/graph_pipeline.py — LangGraph Swarm**

  Orchestrates 3-node stateful swarm: Ingestion Node (math extraction), Threat Analysis Node (ChromaDB + Gemini 2.5 LLM reasoning), and Response Node (triage matrix evaluation).

- **core/math_engine.py — Mathematical Feature Engine**

  Calculates Shannon Entropy H(X) for stolen token detection (>3.85 bits/char) and Haversine Travel Velocity (km/h) for impossible travel detection (>800 km/h).

- **core/vector_store.py — ChromaDB Threat Memory**

  Pre-seeded local vector database matching incoming anomalies against MITRE ATT&CK; techniques (T1539 AiTM, T1550.004 Token Theft, T1110 ATO, T1090 Proxy) using Cosine Similarity.

- **core/mitigation_engine.py — Enterprise SOAR Enforcer**

  Executes sub-second automated containment: session token revocation, perimeter firewall IP block, and real SMTP TLS email alerts to recipient inboxes.

- **tools/real_dataset_streamer.py — 31M Dataset Streamer**

  High-performance O(1) memory generator streaming 1 GB+ Kaggle RBA dataset CSV files with ~35 MB RAM footprint and auto directory resolution.

- **static/index.html — Command Theater HUD**

  Interactive dashboard featuring Central Lock Theater, Tactical World Map Radar, Shannon Entropy Waveform, Multi-Agent Swarm Inspector, and HITL Review Portal.

4. Capability Verification & Objective Matrix

Capability	Module / File	Mechanism & Status
Real-Time Data Streaming	tools/real_dataset_streamer.py	O(1) memory generator over 31M CSVs (PASS)
Impossible Travel Detection	core/math_engine.py	Haversine distance formula >800 km/h (PASS)
Token Theft Detection	core/math_engine.py	Shannon Entropy formula >3.85 bits/char (PASS)
Multi-Agent Orchestration	agents/graph_pipeline.py	LangGraph 3-node stateful AgentGraphState (PASS)
Cognitive Threat Analysis	agents/graph_pipeline.py	Google Gemini 2.5 LLM + ChromaDB RAG (PASS)
Automated Response (SOAR)	core/mitigation_engine.py	Session revoke + IP block + SMTP email (PASS)
Human-In-The-Loop Review	server.py & static/index.html	Analyst review modal for scores 50-80 (PASS)